A Deep Learning Framework for Multilingual Medicinal Plant Identification and Information Retrieval System

Kowsik S, Sourish Cidambaram B G, Selva Subramanian A, Nagarjun M, Dr. R. Baskaran, and Dr. T. Sree Sharmila

[1] 4th Year Students, Department of Computer Science and Engineering
Anna University, Chennai , Tamil Nadu, India
[2] Professor, Department of Computer Science and Engineering
Anna University, Chennai , Tamil Nadu, India
[3] Associate Professor, Department of Computer Science and Engineering
Anna University, Chennai , Tamil Nadu, India

Abstract. Medicinal plants play a vital role in traditional healthcare systems, especially in countries with rich biodiversity and multilingual populations. However, accessing accurate plant-related information remains challenging due to language barriers, unstructured data sources, and the limited availability of intelligent retrieval systems. This paper presents an AI-driven multilingual medicinal plant identification and information system that enables users to query plant-related knowledge using both text and speech inputs in multiple Indian languages.

The proposed system integrates language detection using fastText, neural machine translation through the NLLB-200 model, and semantic information retrieval using sentence-level embeddings. User queries are normalized and mapped to a unified English representation, enabling accurate semantic matching against a curated medicinal plant database. For speech-based interaction, an automatic speech recognition pipeline is employed to convert audio queries into text prior to language processing.

The system is evaluated using standard information retrieval and speech recognition metrics, including accuracy, precision, recall, F1-score, word error rate, and translation quality scores. Experimental results demonstrate that the proposed approach effectively handles multilingual and noisy inputs while maintaining reliable plant identification and information retrieval performance. The system aims to improve accessibility to medicinal plant knowledge for a wider audience, including non-English-speaking users.

Keywords: Multilingual Information Retrieval · Medicinal Plants · Natural Language Processing (NLP) · Neural Machine Translation · Semantic Search · Speech Recognition · NLLB-200 Distilled 600M (No Language Left Behind)

1 Introduction

Medicinal plants have been used for centuries as a primary source of healthcare in many parts of the world. Traditional systems such as Ayurveda, Siddha, Unani, and folk medicine rely heavily on plant-based remedies for disease prevention and treatment. India, being one of the most biologically diverse countries, hosts thousands of medicinal plant species with well-documented therapeutic properties. However, this knowledge is distributed across different languages, regions, and cultural practices, often preserved through oral traditions or region-specific texts.

In the digital era, the preservation and dissemination of medicinal plant knowledge require intelligent information systems that can overcome linguistic and technical barriers. Conventional search systems depend on keyword matching and are often ineffective when users do not know the exact botanical or English names of plants. This problem becomes more severe in multilingual societies where users may refer to the same plant using different regional names or expressions.

Recent advances in artificial intelligence, particularly in natural language processing and deep learning, have enabled the development of multilingual and multimodal systems. Language identification, neural machine translation, speech recognition, and semantic search models have matured to a level where they can be integrated into real-world applications. However, their combined use in the medicinal plant domain remains limited.

This project aims to design an AI-driven system capable of understanding user queries in multiple languages and formats, including speech, and retrieving accurate medicinal plant information. By integrating language detection, translation, and semantic similarity-based retrieval, the system reduces dependency on exact keyword matches and supports natural user interactions. The goal is to create an accessible, scalable, and reliable platform that enhances public access to medicinal plant knowledge.

2 Literature Survey

Multilingual natural language processing has gained significant attention due to the growing demand for inclusive and accessible artificial intelligence systems capable of supporting diverse linguistic communities. Wang et al. [1] explored recent advancements in interactive machine translation systems enhanced with large language models, focusing on how human-in-the-loop mechanisms improve contextual understanding and translation adaptability. Their study demonstrated that LLM-based systems can dynamically incorporate user feedback, refine translations in real time, and preserve semantic consistency more effectively than conventional neural machine translation approaches. The authors further emphasized the importance of maintaining semantic fidelity when translating domain-specific content such as medical and botanical terminology, where even minor errors can result in misinformation. These findings are particularly relevant for

multilingual medicinal knowledge systems that require high accuracy and contextual awareness.

Addressing the challenges faced by low-resource languages, Liang et al. [2] proposed language-specific fine-tuning strategies using Low-Rank Adaptation (LoRA) to enhance machine translation performance. Their approach selectively updates a limited number of parameters in large pretrained models, significantly reducing computational costs while achieving notable improvements in translation quality. Experimental results showed consistent gains in BLEU and chrF scores across multiple low-resource language pairs. This work is especially relevant in the Indian context, where many regional languages lack large annotated corpora. By enabling efficient adaptation with minimal data, the proposed method provides a practical solution for developing scalable and inclusive multilingual translation systems.

Speech-based interaction has become an essential component of modern information systems, particularly for users who prefer voice-based access or face literacy constraints. Porjazovski et al. [3] conducted a comprehensive evaluation of speech embedding techniques, comparing traditional handcrafted acoustic features with modern deep learning and transformer-based representations. Their findings revealed that self-supervised and transformer-based speech embeddings consistently outperform conventional methods across multilingual speech recognition and retrieval tasks. The study also highlighted the influence of pooling strategies and normalization techniques on embedding quality. These insights support the adoption of advanced speech recognition models for processing regional language audio queries in real-world applications.

Extending research in speech recognition, Sharrab et al. [4] presented a systematic review of transformer-based automatic speech recognition systems, emphasizing their robustness, scalability, and adaptability. The authors examined a wide range of models, including encoder–decoder transformers, conformer architectures, and large-scale multilingual ASR systems such as Whisper. Their analysis showed that these models achieve strong performance across diverse languages and acoustic conditions, making them suitable for deployment in practical multilingual environments. Despite these advances, the study also identified challenges related to computational complexity, real-time processing, and bias in speech datasets, highlighting areas for future research.

Semantic search has emerged as an effective alternative to traditional keyword-based retrieval systems. Weckmüller et al. [5] proposed an embedding-based multilingual semantic search framework that enables language-independent retrieval of semantically related content. By leveraging sentence-level embeddings, their approach overcomes the limitations of exact keyword matching and improves retrieval accuracy in multilingual datasets. The authors demonstrated the effectiveness of their framework through urban data case studies, showing improved semantic coherence and cross-lingual retrieval. This work directly informs the design of semantic retrieval mechanisms used in the present system, where user queries must be mapped to relevant medicinal plant information regardless of language or phrasing.

Beyond retrieval, data integration remains a critical challenge in heterogeneous and multilingual environments. Hong and Park [6] surveyed the application of large language models in semantic join tasks, highlighting how embedding-based similarity computation enables reliable linking of heterogeneous textual records. Their review showed that LLM-based semantic joins capture latent relationships that rigid rule-based or schema-driven methods often miss. However, the authors also discussed challenges such as hallucination risks and scalability concerns, emphasizing the need for verification mechanisms and constraint-based validation. These insights are particularly relevant when mapping user queries to structured medicinal plant records.

From a domain-specific perspective, Zouine et al. [7] conducted a comprehensive review of medicinal plant extracts and their antibacterial properties, analyzing factors such as extraction methods, phytochemical composition, and biological mechanisms. Their study highlighted the therapeutic potential of plant-derived compounds while stressing the importance of accurate plant identification and reliable documentation. The authors noted that inconsistencies in naming conventions and incomplete validation studies can hinder clinical translation. This review underscores the societal and scientific importance of building trustworthy medicinal plant information systems that integrate botanical accuracy with validated pharmacological knowledge.

From the perspective of speech-based interaction, Ahlawat and Verma [8] presented a comprehensive survey of deep learning–based automatic speech recognition systems, highlighting the transition from traditional acoustic models to transformer-based architectures and emphasizing their robustness in multilingual and noisy environments.

Focusing on end-to-end speech processing, Feng et al. [9] proposed an adaptive multitask learning framework that jointly models speech recognition and machine translation, demonstrating improved performance for multilingual speech-to-text translation tasks by reducing error propagation across components.

From the viewpoint of multilingual language understanding, Qin et al. [10] conducted an extensive survey of multilingual large language models, analyzing their architectures, training strategies, and cross-lingual representation capabilities, and emphasizing their applicability to tasks such as translation, semantic search, and cross-lingual information retrieval.

Addressing cross-lingual retrieval challenges, Patel et al. [11] investigated information retrieval techniques for low-resource Indian languages, highlighting issues such as transliteration variability and data sparsity, and demonstrating the effectiveness of semantic embedding-based approaches for improving retrieval accuracy.

From a domain-specific healthcare perspective, Iranzo-Sánchez et al. [12] explored the integration of speech recognition and machine translation in multilingual medical information systems, emphasizing the importance of domain adaptation to preserve terminology accuracy and semantic intent.

In the context of semantic information retrieval, Wang et al. [13] proposed a sentence embedding–based retrieval framework for multilingual environments,

demonstrating that dense semantic representations outperform traditional keyword-based methods when handling paraphrased and linguistically diverse queries.

Focusing on Indic language processing, Kumar and Sinha [14] reviewed recent advancements in natural language processing techniques for Indian languages, discussing challenges such as morphological richness, code-mixing, and limited annotated data, and highlighting the role of multilingual pretraining and transfer learning.

From the perspective of multilingual semantic search, Zhang and Huang [15] introduced a transformer-based embedding framework that enables language-agnostic semantic retrieval, achieving improved performance across languages without relying on explicit machine translation.

3 Methodology

The proposed system follows a modular and layered architecture designed to efficiently process multilingual text and audio queries for medicinal plant information retrieval. Each module in the system performs a clearly defined function, enabling scalability, maintainability, and easy integration of additional languages or models in the future. The overall methodology consists of a sequence of interconnected components that together form an end-to-end information retrieval pipeline, beginning with user query acquisition and ending with semantic search–based result generation.

Users interact with the system through either text-based or audio-based input modes. Text queries can be entered in English or any supported Indian regional language, allowing users to express queries in their preferred linguistic form. Audio queries are captured through a microphone interface and are particularly beneficial for users who may have limited typing skills or literacy constraints.

This multimodal query acquisition strategy significantly enhances accessibility and usability, ensuring that the system can be effectively used by a diverse user base.

For textual inputs, the system applies a FastText-based language identification model to automatically determine the source language of the query. FastText is selected due to its lightweight architecture, fast inference time, and strong performance in identifying short and noisy text inputs across multiple languages.

3.1 Module 1: Multilingual Query Acquisition and Semantic Retrieval

Module 1 constitutes the foundational linguistic and semantic processing layer of the proposed AI-Based Multilingual Plant Information System. Its primary objective is to accurately interpret user queries provided in diverse Indian languages and modalities, normalize them into a unified representation, and retrieve the most relevant medicinal plant information from the textual database. This module plays a critical role in ensuring that the system remains accessible, language-agnostic, and robust to variations in user input, including spelling

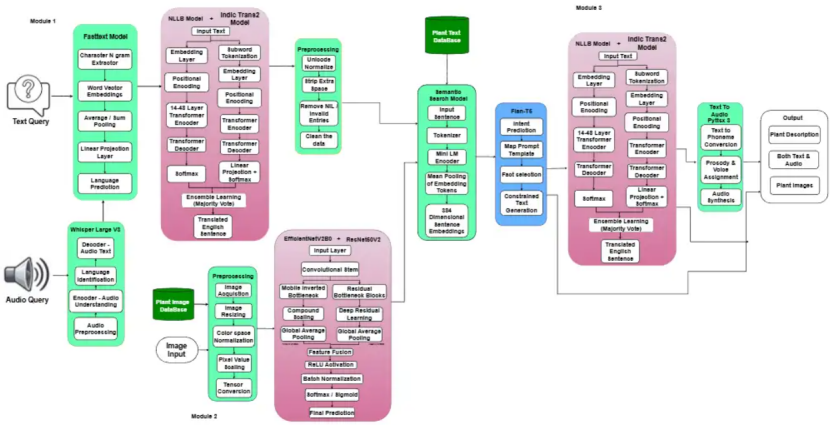

Fig. 1. Overall architecture of the proposed system highlighting Module 1

differences, script diversity, and speech-based queries. By integrating language detection, preprocessing, speech recognition, machine translation, and hybrid semantic retrieval, Module 1 enables reliable interaction between users and the knowledge base.

The processing pipeline begins with automatic language detection applied to the raw user input. For text-based input, a fastText-based language identification model is employed, leveraging character-level n-gram representations to effectively handle short and noisy queries. The model extracts character n-grams, generates embeddings, applies pooling, and predicts the most likely language code. Confidence thresholds and Unicode script heuristics are used to enhance robustness.

Following language identification, the input undergoes preprocessing and normalization. Unicode normalization using the NFC standard is applied, extra whitespace and invalid entries are removed, and numerals are standardized. For Indic languages, transliteration into a Latin-based script is performed to improve matching against database entries.

For audio queries, automatic speech recognition is performed using the Whisper Large v3 Turbo model, which detects the spoken language and transcribes speech into text. The transcription is passed through the same normalization pipeline to maintain consistency.

All normalized queries are translated into English using the Indic Translation 2 model and NLLB (No Language Left Behind) (hybrid) to enable unified semantic indexing. Hybrid retrieval is then performed using exact matching, fuzzy matching, and MiniLM-based dense semantic search. The final response is formatted in a structured, intent-aware manner.

Overall, Module 1 provides a robust multilingual query understanding and semantic retrieval framework, enabling accurate access to medicinal plant knowledge across languages and modalities.

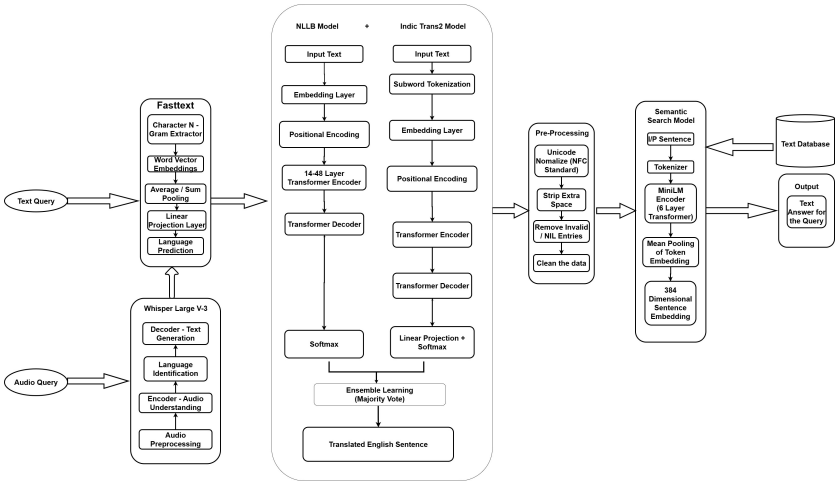

Fig. 2. Multilingual Query Acquisition and Semantic Retrieval in Module 1

3.2 Module 2: Vision–Language-Based Plant Identification and Semantic Retrieval

3.3 Hybrid Vision–Language Module for Plant Identification

The proposed module implements a hybrid vision–language framework for accurate plant identification and semantic information retrieval. The system integrates a hybrid convolutional neural network composed of EfficientNet-B0 and ResNet50V2 with a transformer-based semantic search model to jointly leverage visual and textual representations.

The pipeline begins with plant image acquisition from either a live camera feed or an image repository. To ensure robustness across varying illumination conditions, backgrounds, and image resolutions, the input image undergoes a standardized pre-processing pipeline. This includes resizing to a fixed spatial resolution, color normalization to reduce illumination variance, pixel value scaling, and conversion into a tensor representation compatible with deep learning frameworks.

Following pre-processing, the image tensor is passed through the hybrid feature extraction backbone. EfficientNet-B0 is employed for its computational efficiency and ability to capture fine-grained visual features such as leaf texture, venation patterns, and color distributions. In parallel, ResNet50V2 extracts deeper semantic and structural features through residual learning. The feature maps from both networks are subjected to global average pooling and concatenated to form a unified visual embedding that represents the semantic content of the plant image.

To align visual information with textual knowledge, the fused visual embedding is mapped into a shared semantic space using a transformer-based language model. A MiniLM encoder with six transformer layers is utilized to gen-

erate contextualized text embeddings from plant-related descriptions stored in a structured knowledge base. Semantic similarity between visual and textual embeddings is computed using cosine similarity to retrieve the most relevant plant descriptions.

The final output presents the predicted plant identity along with detailed semantic information, enabling both accurate classification and explainable retrieval.

3.4 Module 3: Intent-Aware Response Generation and Multilingual Output Delivery

This module implements a structured natural language processing pipeline designed to interpret user queries, generate accurate plant-related responses, and deliver multilingual outputs in a controlled and semantically consistent manner. The process begins with the user's input query, which is first analyzed by the intent prediction component. In this stage, keyword extraction is performed to identify salient terms related to plant species, characteristics, or usage. These extracted keywords are then mapped to predefined intent categories, allowing the system to classify the user's query into a labeled intent that determines the nature of the requested information, such as plant identification, medicinal properties, or general botanical facts. This intent-aware classification ensures that subsequent processing stages operate with a clear understanding of the user's informational needs.

Once the user intent is identified, the system proceeds to the prompt construction stage, which is responsible for generating a structured and contextually rich prompt for the language model. Relevant plant facts are retrieved from a curated knowledge source based on the predicted intent and extracted keywords. These facts are then mapped to predefined prompt templates, ensuring consistency and alignment with the expected response format. The selected factual information is injected into the template to form a structured prompt that constrains the language model to generate accurate, relevant, and non-hallucinatory responses. This structured prompt design plays a critical role in maintaining factual correctness while enabling flexibility in natural language generation.

The structured prompt is subsequently processed by a text generation module based on the Flan-T5 model. In this stage, prompt parsing is performed to interpret the structure and constraints embedded in the prompt. The model selects the most relevant factual elements and performs constrained text generation, producing coherent and context-aware responses that adhere strictly to the provided factual content. By employing constrained generation rather than free-form text synthesis, the system minimizes ambiguity and enhances the reliability and interpretability of the generated output.

To support multilingual accessibility, the generated English response is passed to a translation module based on the Indic Translation 2 model. The translation process involves segmenting the generated text into manageable units, translating each segment into the target language, and recombining the translated

segments to produce a fluent and grammatically correct response. This modular translation strategy ensures semantic consistency while preserving linguistic fluency across diverse Indian languages.

The final output of the module is a translated, user-friendly response that accurately reflects the user's original query intent and delivers reliable plant-related information. By integrating intent prediction, structured prompt construction, constrained text generation, and multilingual translation, Module 3 provides an efficient, scalable, and interpretable natural language interface suitable for intelligent plant information systems, educational platforms, and multilingual user environments.

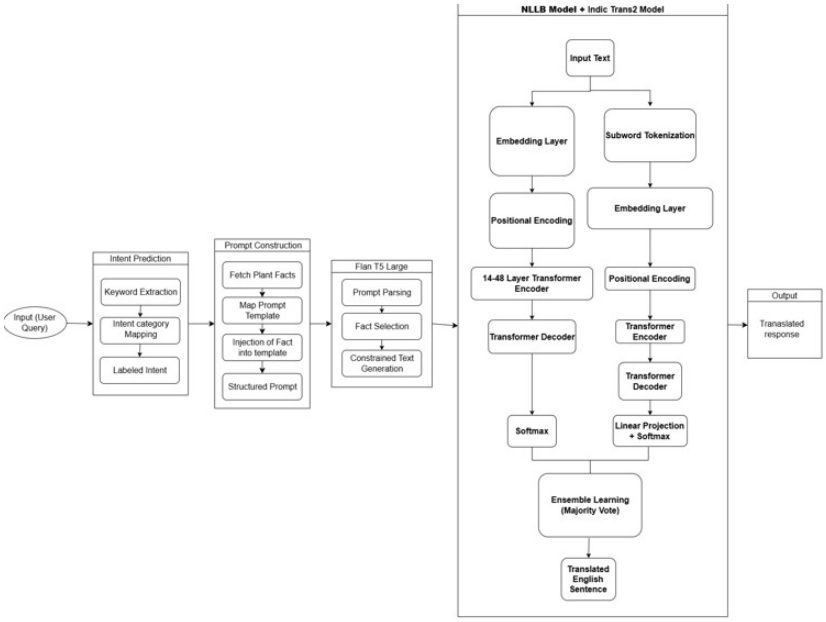

Fig. 3. Intent-aware response generation and multilingual output workflow in Module 3

4 Dataset Description

The dataset used in this study consists of a carefully curated collection of medicinal plant records compiled from authenticated and reliable sources. Each plant entry is uniquely identified and includes its botanical (scientific) name, which serves as the primary reference, along with the commonly used English name. To support multilingual access and regional usability, vernacular names are provided in multiple Indian languages, including Tamil, Telugu, Hindi, Malayalam, and Sanskrit. This multilingual representation enables the system to handle user

queries expressed in diverse linguistic forms while maintaining consistency across plant records.

In addition to naming information, each dataset record contains rich descriptive attributes that capture both botanical and medicinal knowledge. These attributes include detailed botanical descriptions outlining morphological characteristics, traditional medicinal uses documented in classical and contemporary sources, pharmacological properties associated with therapeutic effects, and a list of active constituents responsible for the plant's biological activity. Collectively, these fields provide comprehensive semantic context that is essential for accurate information retrieval and interpretation.

Prior to use in the retrieval pipeline, the textual data underwent systematic cleaning and preprocessing. This process involved removing null or incomplete entries, standardizing naming conventions across languages, and normalizing textual content to ensure uniform representation. Variations in spelling, capitalization, and formatting were resolved to reduce noise and prevent inconsistencies that could negatively impact embedding generation and similarity computation. Such preprocessing ensures that semantically equivalent content is treated consistently by downstream natural language models.

For semantic retrieval, relevant descriptive fields—particularly those related to medicinal usage, pharmacological activity, and plant characteristics—were concatenated to form a unified textual representation for each plant. These consolidated texts were then encoded into dense vector representations using pretrained sentence embedding models. The resulting embeddings constitute the system's semantic knowledge base, enabling efficient similarity-based retrieval that goes beyond exact keyword matching.

To evaluate the speech-based interaction component, a small but representative set of audio queries was collected in selected regional languages. These recordings were used to assess the performance of the automatic speech recognition module and to analyze how transcription quality influences subsequent translation and retrieval accuracy. By incorporating both textual and audio inputs into the evaluation process, the dataset supports a comprehensive assessment of the system's multilingual and multimodal capabilities.

5 Results

This section presents the experimental results obtained from the proposed AI-driven multilingual medicinal plant identification and information system and provides a detailed discussion of its performance. The evaluation focuses on text-based queries, audio-based queries, multilingual robustness, and the effectiveness of semantic search in handling user variations and ambiguities.

5.1 Text Query Evaluation

The text-based evaluation was conducted using queries formulated in multiple Indian languages, including Tamil, Hindi, Telugu, Malayalam, and English. Each

query was designed to retrieve information related to medicinal plant properties, such as benefits, usage, and local names. The system successfully detected the input language using the fastText language identification model and translated non-English queries into English using the NLLB-200 multilingual translation model.

The evaluation metric for text queries was binary correctness, where a query was marked correct if the system retrieved the expected plant entry from the dataset. The results demonstrated that the proposed system achieved reliable performance for most common medicinal plant queries, particularly when the plant name appeared explicitly or implicitly in the user input. The use of fuzzy string matching and semantic similarity scoring significantly improved retrieval accuracy for queries containing spelling variations, transliteration differences, or incomplete plant names.

However, challenges were observed in cases where the query lacked explicit plant references and relied solely on symptom-based descriptions. In such scenarios, semantic similarity occasionally matched related but incorrect plant entries. This behavior highlights the inherent ambiguity present in natural language queries and the dependence of semantic retrieval on dataset coverage and contextual representation.

5.2 Audio Query Evaluation

For audio-based queries, speech input was captured through the user interface and processed using the Whisper Large V3 automatic speech recognition model. The recognized text was then forwarded to the same multilingual text processing pipeline used for text queries. The audio evaluation emphasized transcription accuracy and its impact on downstream retrieval performance.

Experimental observations indicated that accurate speech transcription directly influenced plant identification correctness. Queries recorded in clear acoustic conditions with minimal background noise yielded highly reliable transcriptions, resulting in correct plant retrieval. Conversely, transcription errors caused by pronunciation variations, background noise, or dialectal differences occasionally led to incorrect plant identification.

The Word Error Rate (WER) of the ASR component played a critical role in overall system performance. Higher WER values were correlated with reduced retrieval accuracy, demonstrating the cascading effect of speech recognition errors on semantic search and plant matching. Despite these challenges, the Whisper model exhibited strong robustness across multiple Indian languages, making it suitable for real-world multilingual audio query scenarios.

5.3 Multilingual Translation Performance

The NLLB-200 translation model enabled effective conversion of user queries from Indian languages into English, which served as the system's base language for semantic processing. Translation quality was evaluated using both automatic

```
answer_query("வேம்பின் பயன்கள் என்ன?")

{'PLANT_NAME': 'Neem tree',
 'ANSWER_TEXT': 'Usages: i) The young leaves and black pepper is made into a paste and taken internally for Intestinal helminthiasis. ii) A few drops of oil taken in betel-
leaf provide relief in Asthma.'}

answer_query("మేక యొక్క ప్రయోజనాలు ఏమిటి?")

{'PLANT_NAME': 'Neem tree',
 'ANSWER_TEXT': 'Usages: i) The young leaves and black pepper is made into a paste and taken internally for Intestinal helminthiasis. ii) A few drops of oil taken in betel-
leaf provide relief in Asthma.'}

answer_query("Give the poem of Abrus precatorius")

{'PLANT_NAME': 'Wild liquorice',
 'ANSWER_TEXT': 'Tamil poem: நெஞ்சிநோய் பித்தம் நீற்றமுங் கல்காமால வேத்திநோய் பசேஷ்ாப வெப்புடனே – கோத்திட்ட ஐயமுதல் யாவும்போம் ஆவிமையே! காட் டிதுறை
செய்யகுன்றி மின்னிலைதாய்ச் சேர்.'}
```

Fig. 4. Correct Detection of Plant names after detecting the keyword in semantic search model

```
evaluate_single_query("what is the benefits of Dragon fruit?", "dragon Fruit")

{'Query': 'what is the benefits of Dragon fruit?',
 'Expected': 'dragon fruit',
 'Predicted': 'indian atees',
 'Accuracy': 0,
 'Remark': '❌ Wrong',
 'Answer_Text': 'Usages: Powder of tuberous root mixed with honey is given in Cough, Coryza, Fever and Vomiting in children. Decoction of tuberous root is administered in
Intermittent fever.'}
```

Fig. 5. System response when the queried medicinal plant is not found in the database

metrics and manual inspection. The model preserved the semantic intent of most queries, even when literal word-to-word translation was not achieved.

In certain cases, especially for culturally specific or traditional medicinal terms, the translated output differed lexically from the expected phrasing but retained the correct semantic meaning. This behavior did not significantly impact plant retrieval due to the semantic embedding-based search employed in the system. However, rare mistranslations introduced contextual drift, which could lead to suboptimal matching during semantic search.

5.4 Semantic Search and Plant Matching Analysis

Semantic retrieval was implemented using sentence embeddings generated by the Sentence-BERT model (all-MiniLM-L6-v2). This approach allowed the system to capture contextual meaning rather than relying solely on keyword matching. The semantic similarity scores effectively ranked relevant plant entries even when direct string matches were absent.

The integration of semantic search proved particularly beneficial for handling user queries with informal phrasing, incomplete names, or indirect references to plant properties. Additionally, combining semantic similarity with fuzzy string matching helped mitigate false positives and improved overall robustness.

Despite these advantages, semantic search occasionally retrieved closely related but incorrect plant entries when the similarity score threshold was set too low. To address this issue, a minimum similarity threshold was enforced, and queries failing to meet this criterion were correctly labeled as “Plant not found in database.” This strategy reduced incorrect predictions and improved system reliability.

5.5 Overall System Performance

The overall system performance reflects the combined effectiveness of language detection, translation, speech recognition, fuzzy matching, and semantic search. The modular design ensured that each component contributed independently while maintaining interoperability within the pipeline. The system demonstrated strong multilingual capabilities and practical usability for medicinal plant information retrieval.

The results indicate that the proposed approach is well-suited for low-resource language settings and traditional knowledge domains, where spelling variations, transliteration inconsistencies, and speech diversity are common. While the system performs effectively under most conditions, future improvements can focus on enhanced domain-specific fine-tuning, expanded dataset coverage, and improved handling of symptom-only queries.

In summary, the experimental results validate the feasibility and effectiveness of the proposed AI-driven multilingual medicinal plant identification system. The discussion highlights both strengths and limitations, providing insights for further research and real-world deployment.

6 Performance Metrics

Although the proposed system operates as a semantic retrieval framework, its behavior can be equivalently modeled as a single-label classification task, where each user query is expected to map to exactly one correct medicinal plant from the database. Based on this formulation, standard classification and component-wise evaluation metrics are employed to assess system performance.

6.1 Retrieval Performance Metrics

For evaluating retrieval accuracy, the following outcomes are defined:

- **True Positive (TP):** The system retrieves the correct medicinal plant for a given query.
- **False Positive (FP):** The system retrieves an incorrect plant.
- **False Negative (FN):** The system fails to retrieve the correct plant, either by returning an irrelevant match or indicating that the plant is not found.

Using these definitions, three widely adopted metrics—Precision, Recall, and F1-score—are computed.

Precision Precision measures the reliability of the system's retrieved results and is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

A high precision score indicates that when the system outputs a medicinal plant name, it is likely to be correct. This is especially important in healthcare-related applications, where incorrect retrievals may lead to misinformation.

Recall Recall quantifies the system's ability to successfully retrieve the correct plant when it exists in the database:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

High recall implies that the system rarely misses relevant plant entries, even in the presence of linguistic variation or noisy inputs.

The F1-score represents the harmonic mean of precision and recall:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

The F1-score provides a balanced evaluation, particularly useful when errors arise from mistranslation, fuzzy matching ambiguities, similar plant names (e.g., Neem vs. Margosa), or automatic speech recognition errors.

Together, precision and recall offer insight into whether the system predominantly fails by retrieving incorrect plants or by failing to retrieve any relevant plant at all.

6.2 Speech Recognition Evaluation — Word Error Rate (WER)

For queries submitted through the audio interface, spoken input is first transcribed into text using a transformer-based automatic speech recognition model. Errors at this stage propagate through translation and semantic search modules, directly affecting retrieval accuracy.

The performance of the speech recognition component is evaluated using Word Error Rate (WER), defined as:

$$\text{WER} = \frac{S + D + I}{N} \tag{4}$$

where:

- S denotes the number of word substitutions,
- D denotes the number of word deletions,
- I denotes the number of word insertions, and
- N represents the total number of words in the reference transcription.

For example, if a user utters the Tamil query "Vembin payanugal enna" and the ASR model produces a distorted transcription with missing or substituted words, the WER value increases. A high WER indicates poor transcription quality, which in turn degrades translation accuracy and semantic retrieval performance.

WER is therefore a critical metric for quantifying how much the speech recognition module influences end-to-end system reliability.

As shown in Fig. 6, Evaluation Metrics.

F1-Score

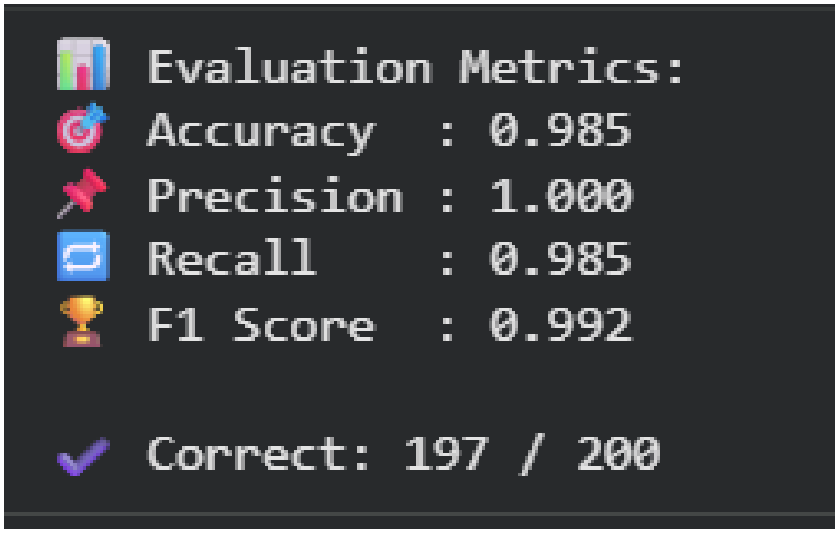

Fig. 6. Evaluation metrics results

6.3 Machine Translation Evaluation — BLEU and chrF

For multilingual text and speech-derived queries, the system relies on neural machine translation to convert the input into English before semantic search. Translation quality directly affects the preservation of plant names and user intent.

BLEU Score The Bilingual Evaluation Understudy (BLEU) score measures n-gram overlap between the model-generated translation and a reference translation. It is defined as:

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{5}$$

BLEU is effective for evaluating longer, structured sentences, though its sensitivity decreases for short or query-style inputs.

chrF Score The chrF metric evaluates translation quality using character-level n-gram precision and recall. It is particularly well suited for morphologically rich and agglutinative languages such as Tamil, Telugu, and Malayalam. chrF often correlates more closely with human judgment for short sentences and single-word queries, making it a valuable complement to BLEU.

As shown in Fig. 8, the proposed system achieves consistent accuracy across multilingual queries.

	Query	Predicted Plant	Expected Plant	Remark
1	What is the benefits of Indian gum-arabic tree?	Indian gum-arabic tree	Indian gum-arabic tree	Correct
2	What is the benefits of Greater cardamom?	Greater cardamom	Greater cardamom	Correct
3	What is the benefits of Alternanthera?	Alternanthera	Alternanthera	Correct
4	What is the benefits of Kocco?	Kocco	Kocco	Correct
5	What is the benefits of The spiny bamboo?	The spiny bamboo	The spiny bamboo	Correct
...	...	...	...	...
196	What is the benefits of Rose wood?	Rose wood	Rose wood	Correct
197	What is the benefits of Andrographis echioidis...		Andrographis echioidis Nees. (Acanthaceae)	Wrong
198	What is the benefits of Velvette-leaf pareira?	Velvette-leaf pareira	Velvette-leaf pareira	Correct
199	What is the benefits of Alexandrian Laurel?	Alexandrian Laurel	Alexandrian Laurel	Correct
200	What is the benefits of Ivy gourd?	Ivy gourd	Ivy gourd	Correct

200 rows × 4 columns

Fig. 7. Accuracy performance for different plant names

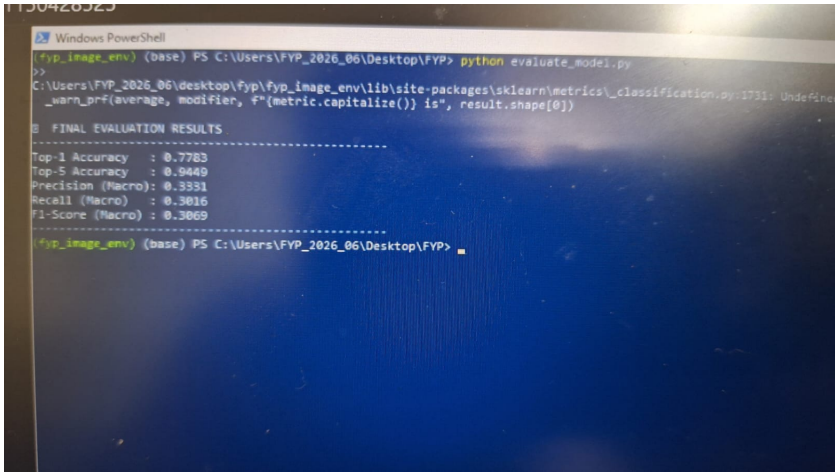

Fig. 8. Mod 2 evaluation metrics and output respectively.

6.4 Human Evaluation

Given the domain-specific nature of medicinal plant terminology, automated metrics alone are insufficient. Therefore, human spot checks are conducted to validate translation and retrieval outputs. Evaluators verify whether:

- Medicinal plant names are preserved correctly across translation,
- The intended usage or benefit is accurately conveyed, and
- No semantic drift occurs (e.g., mapping “veppilai” to an unrelated plant such as taro leaves).

Human evaluation ensures domain fidelity and safeguards against errors that automated metrics may fail to detect.

7 Conclusion

This work presented a multilingual and multimodal information retrieval system designed to provide accurate medicinal plant information through both text and speech-based queries. By integrating language detection, automatic speech recognition, neural machine translation, and semantic search into a unified pipeline, the proposed system effectively addresses the challenges posed by linguistic diversity and varied user interaction modes.

The use of transformer-based models for speech recognition and sentence-level embeddings for semantic retrieval enables the system to move beyond traditional keyword-based approaches and retrieve plant records based on conceptual similarity. Translation into a common base language ensures consistent semantic processing across regional languages, while threshold-based validation helps prevent incorrect or misleading outputs. Experimental evaluation using precision, recall, F1-score, word error rate, and translation metrics demonstrates that the system performs reliably under multilingual and noisy input conditions.

The results highlight the feasibility of deploying inclusive AI-driven retrieval systems for traditional medicinal knowledge, particularly in contexts where users rely on vernacular languages or voice-based interaction. By leveraging authenticated plant data and modern natural language processing techniques, the system contributes toward building trustworthy and accessible digital healthcare support tools.

Future work will focus on expanding the dataset to include a broader range of medicinal plants, improving robustness for code-mixed speech, and incorporating multimodal inputs such as plant images for enhanced identification. Additionally, integrating domain-specific fine-tuning and real-time deployment optimizations may further improve system performance and usability in practical settings.

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